

WW8L CODEPLUG MAGIC

One Programmer. One Database. Every Radio.

Keep every repeater, list and radio setting in one place, and program all your radios from it: AnyTone, Yaesu, Icom, Kenwood, Baofeng and more, over USB, a programming cable or the radio's own microSD card.

User guide · for Codeplug Magic v26.10.16

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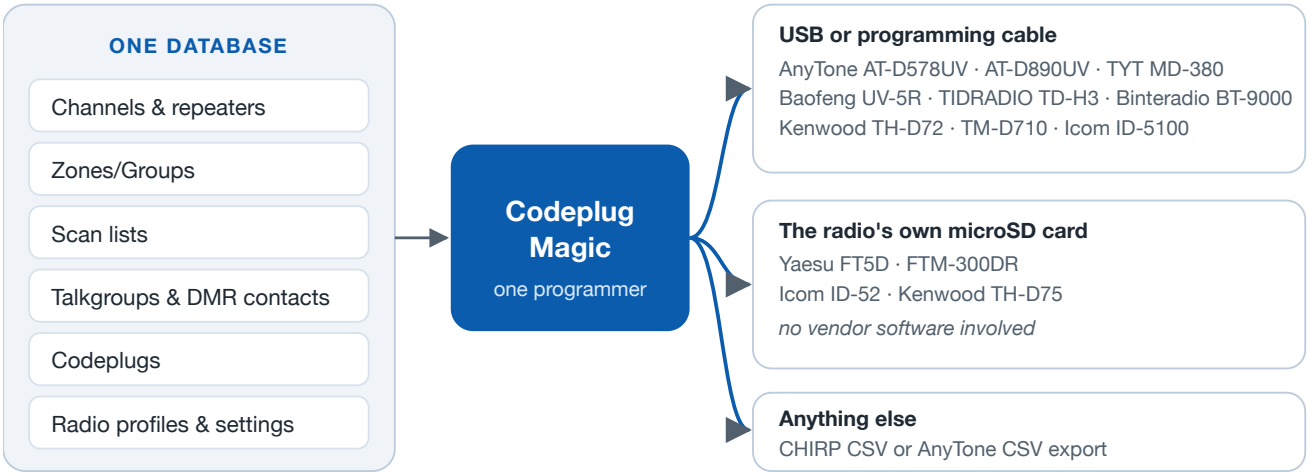
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CHAPTER 1

One programmer for every radio

Every radio maker ships its own programming software, with its own file format, its own channel editor and its own idea of a zone. Own four radios and you keep four copies of the same repeaters, in four programs, and fix every typo four times. Codeplug Magic replaces that shelf of programs with one.



What one database buys you

- **Enter a repeater once.** Add it by hand, import it from RepeaterBook or a CSV file, and it's available to every radio you own. Fix a tone once and every radio gets the fix the next time you program it.
- **Every radio's settings in one place.** Each radio gets a profile holding its menu settings (display, sounds, APRS, DMR ID…), edited with the same screen whatever the brand.
- **The same steps for every radio.** Pick a codeplug, click **Program Radio**. Whether the radio wants a USB cable, a serial cable or its microSD card, the workflow doesn't change.
- **The app knows each radio's limits.** Modes, transmit and receive bands, name length, memory count, zones and banks are built in, so it never writes something a radio can't use, and it tells you what it left out and why.
- **One file to back up. Settings → Database → Back up database…** saves every channel, list, codeplug, radio profile, talkgroup and DMR contact in a single file. Radios are backed up before every cable write.
- **One codeplug, many radios.** A codeplug isn't locked to one radio: switch its radio profile and the same channels program into another (Chapter 7).

CHAPTER 2

The building blocks

Four things, each with one job. Knowing which one owns what makes the rest of this guide obvious.

PIECE	WHERE IN THE APP	WHAT IT OWNS
Channel	Channels	One frequency pair with its mode, tones or DMR settings, and both a long and a short name. Shared by every codeplug and every radio.
Channel list	Zones/Groups	An ordered group of channels. On the radio it becomes a zone, bank or memory group, depending on what that radio has.
Codeplug	Codeplugs	A name, an *ordered* set of channel lists, any scan lists, and **one** radio profile at a time.
Radio profile	Radio Profiles	One physical radio you own: its model, and that radio's menu settings (display, sounds, APRS, DMR ID…).

Why channels aren't tied to a radio

A channel describes the repeater, not your radio. A DMR repeater is still a DMR repeater when you're holding an analog handheld; the handheld simply can't use it. So Codeplug Magic never asks you to keep a separate “UV-5R channel list”. You keep one library, and each radio takes what it can when you program it.

That's also why a channel carries two names. Radios with room for 16 characters show the long one (“AC0KC Fort Lupto”). A radio that can only show 7 shows the short one (“AC0KC”) rather than a chopped-off stump of the long name.

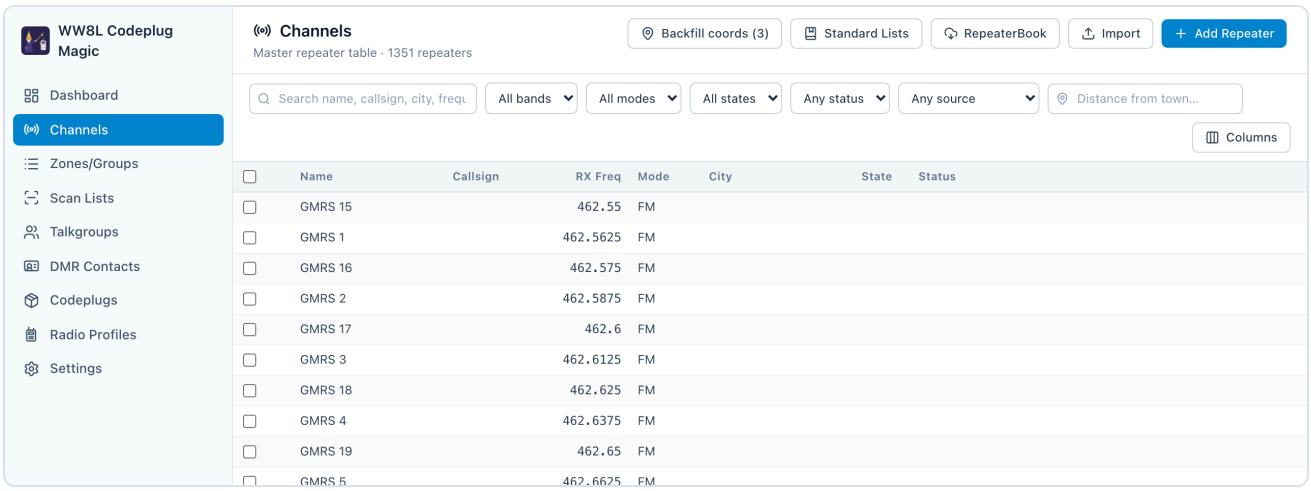
One profile per radio, not per model

If you own two FT5Ds set up differently (one for hiking with APRS beaconing, one for the house), make two profiles. The profile carries that individual radio's menu settings, so the codeplug can stay the same while each radio keeps its own personality.

CHAPTER 3

Your channel library

Everything starts in **Channels**: one table of every repeater and simplex frequency you use, whatever radio it's for. Fill it once from RepeaterBook, a file, or by hand.



The Channels page. Filter by band, mode, state, status or source, or by distance from a town.

Four buttons at the top fill the library:

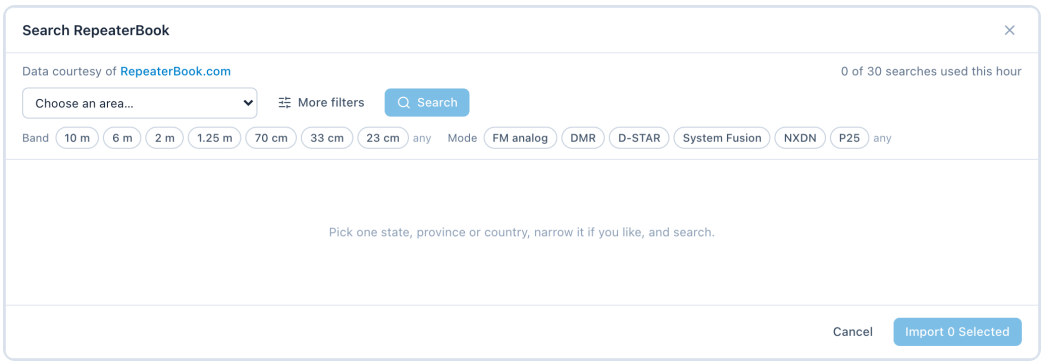
BUTTON	WHAT IT DOES
RepeaterBook	Searches RepeaterBook from inside the app and imports the repeaters you tick.
Import	Reads a file: a RepeaterBook export (CSV or JSON), a CSV from CHIRP or anything else, or a Codeplug Magic channel backup.
Standard Lists	Adds ready-made service channels: GMRS, FRS, MURS, Marine VHF, NOAA Weather and CB.
Add Repeater	Enter one by hand. Type a city and its coordinates are filled in for you.

Search RepeaterBook

Get a free token from RepeaterBook (*API Apps & Tokens* on their site) and paste it into **Settings** → **RepeaterBook**. Then click **RepeaterBook** on the Channels page and do the following:

1. Choose an area: one state, province or country.
2. Narrow the search by band and mode, or open **More filters**.
3. Click **Search**, then tick the repeaters you want.
4. Click **Import Selected**.

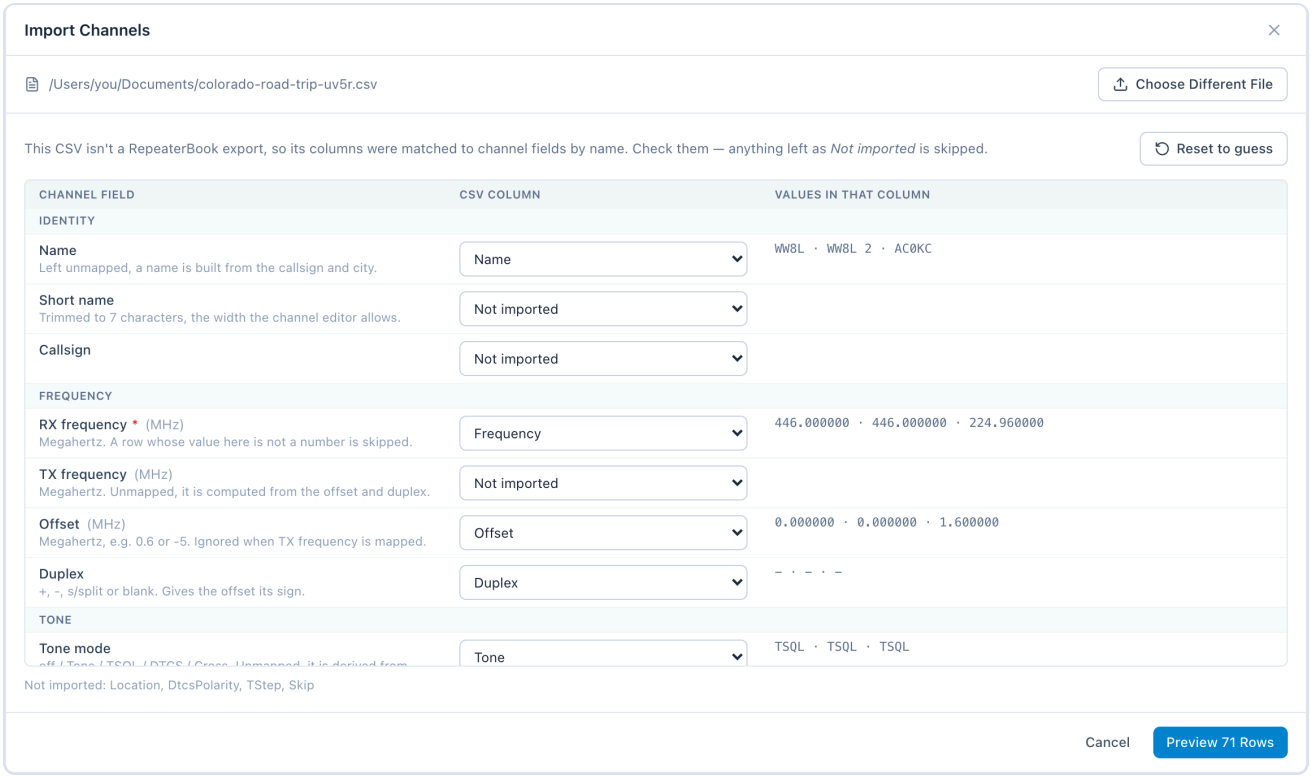
Repeaters already in your library are marked. Importing one again refreshes it instead of making a copy. RepeaterBook limits how often you can search, so the dialog shows how many searches you've used this hour.



Search RepeaterBook: pick an area, narrow it by band and mode, and search.

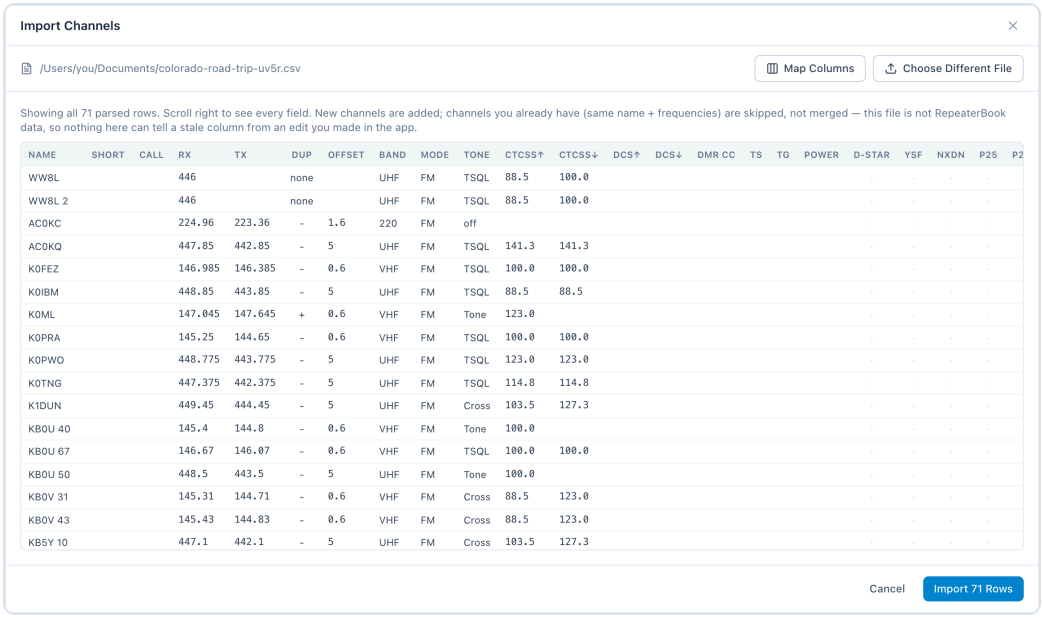
Import a CSV file

A RepeaterBook export is recognised and goes straight to a preview. Any other CSV, from CHIRP for example, opens the column mapper first. The app guesses which column is which and shows you sample values from each, so you can confirm a column really holds frequencies before anything is imported. Anything left as *Not imported* is skipped.



A CHIRP CSV in the column mapper. The right-hand column shows real values from the file.

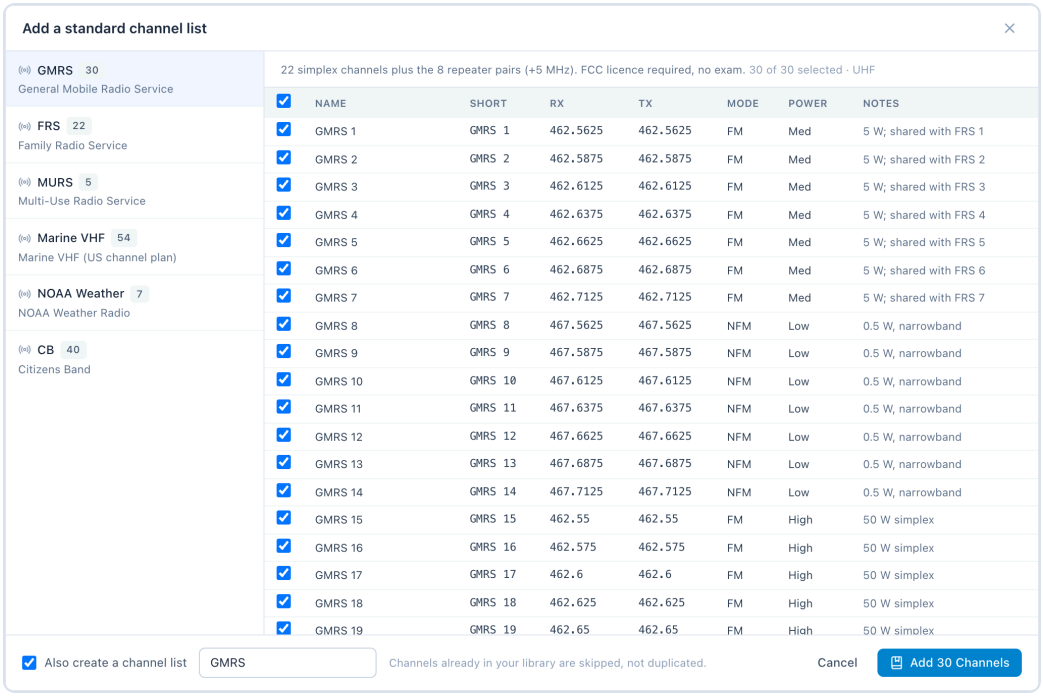
Click **Preview** to see every row as it will be imported, then **Import**. Channels you already have (same name and frequencies) are skipped, not merged, so importing the same file twice does no harm.



The preview: 71 rows from the file, with band, mode and tones already worked out.

Standard lists

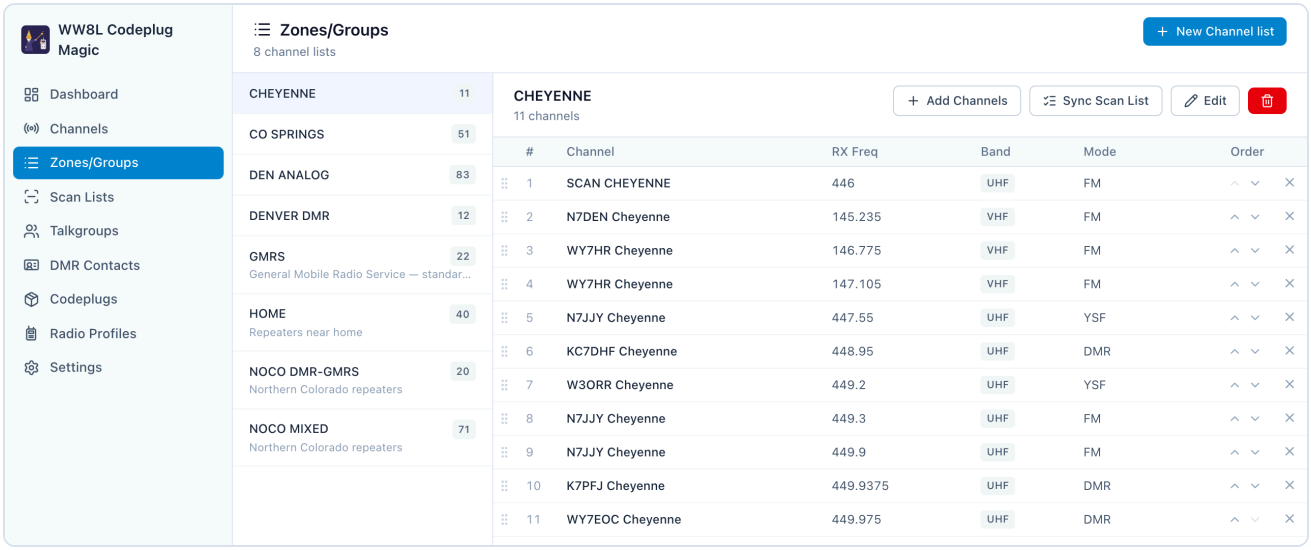
For the services everyone programs the same way, **Standard Lists** adds the whole set in one go. Tick *Also create a channel list* to get a ready-made Zones/Groups list too. Channels already in your library are skipped, not duplicated.



GMRS: 22 simplex channels plus the 8 repeater pairs, with power levels.

Group channels into lists

Zones/Groups holds your channel lists. Select channels on the Channels page and click **Add to channel list**, or open a list and use **Add Channels**. Drag rows to set the order. The order here is the order on the radio.



A channel list mixes modes freely. Each radio takes the channels it can use when you program it.

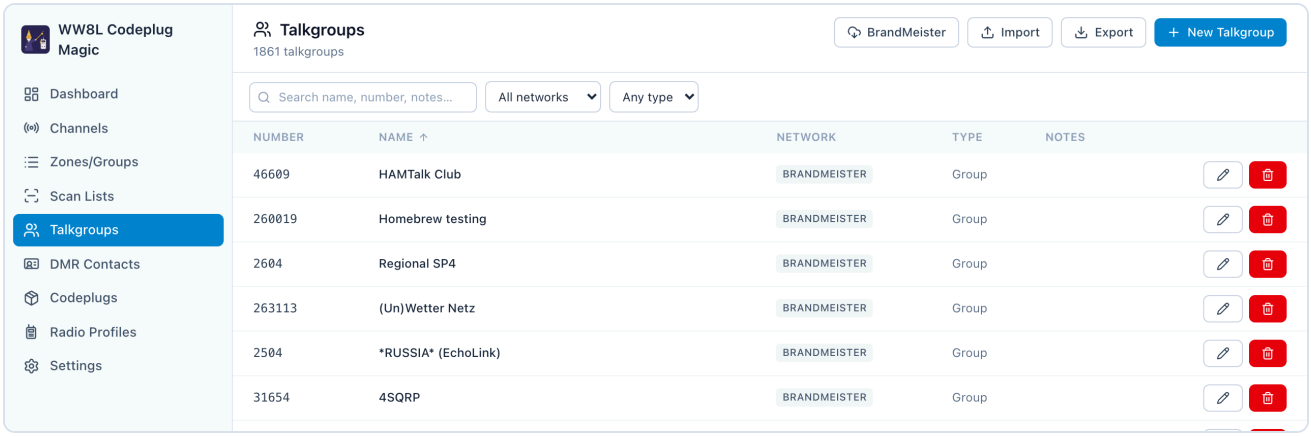
CHAPTER 4

Talkgroups and DMR contacts

DMR needs two more kinds of data: the talkgroups each repeater carries, and the caller-ID list that puts a name on the screen when someone keys up. Both live in the same database and are shared by every DMR radio you own.

Talkgroups

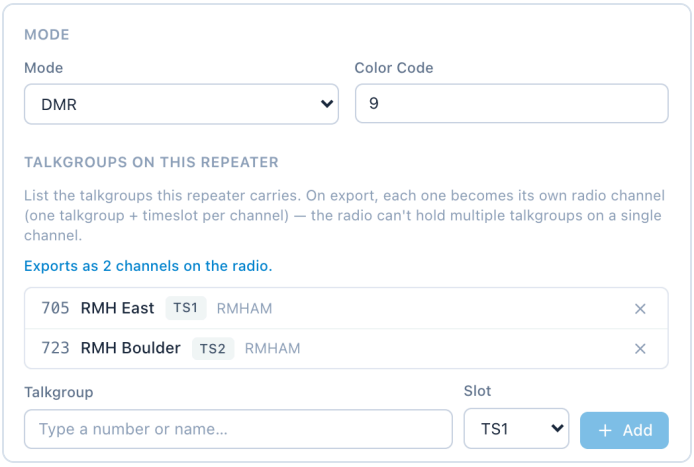
The **Talkgroups** page is your talkgroup library. Click **BrandMeister** to pull the current BrandMeister list. **Import** reads a CSV for other networks, and **New Talkgroup** adds one by hand.



The Talkgroups page, filtered by network and type.

Talkgroups on a repeater

Open a DMR repeater on the Channels page and scroll to *Talkgroups on this repeater*. Add each talkgroup it carries, with its timeslot. A radio can hold only one talkgroup per channel, so each assignment becomes its own channel on the radio, and the panel tells you how many that will be.

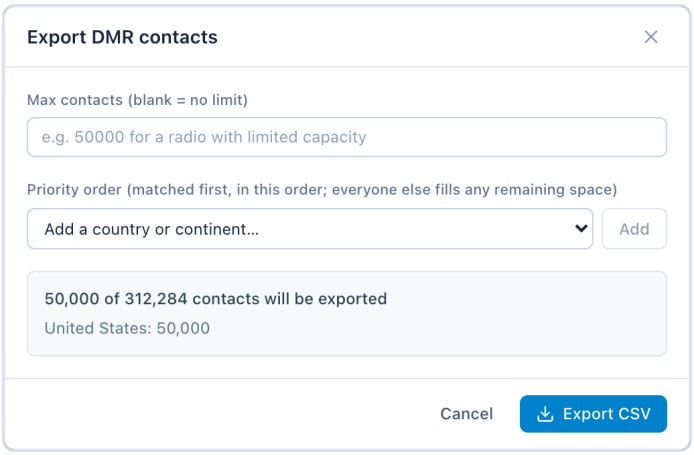


N0SZ Boulder carries RMH East on TS1 and RMH Boulder on TS2, so it exports as 2 channels.

DMR contacts (caller ID)

DMR Contacts keeps a local copy of the RadioID.net user list (over 300,000 operators). Click **Refresh from RadioID.net** now and then to update it. Most radios can't hold the whole list, so you choose what goes in:

- **Directly to the radio:** AnyTone radios have a **Call-sign DB** option in their Program Radio dialog. The TYT MD-380 can fill its spare contacts from the library.
- **As a file: Export…** writes a CSV. Set a maximum and a priority order of countries or continents. The priority matches are written first, and everyone else fills any space that's left.



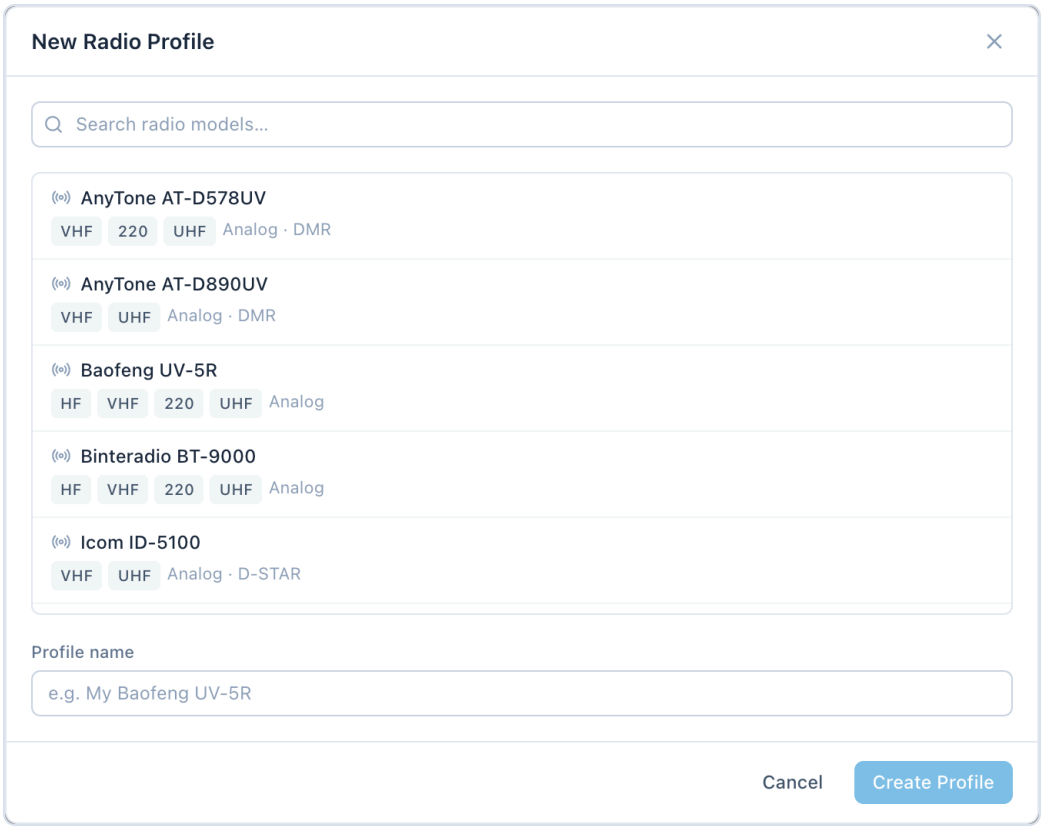
Capped at 50,000 contacts with the United States first: 50,000 of 312,284 will be exported.

CHAPTER 5

Set up each radio once

Make a profile for every radio you'll program. You do this once per radio, not once per codeplug.

1. Open **Radio Profiles** in the sidebar and click **New Profile**.
2. Find your radio in the list. Type in **Search radio models…** to narrow it. Each model shows its bands and modes.
3. Give it a **Profile name** that says which radio it is, for example “Truck mobile (AT-D578UV)”, then click **Create Profile**.



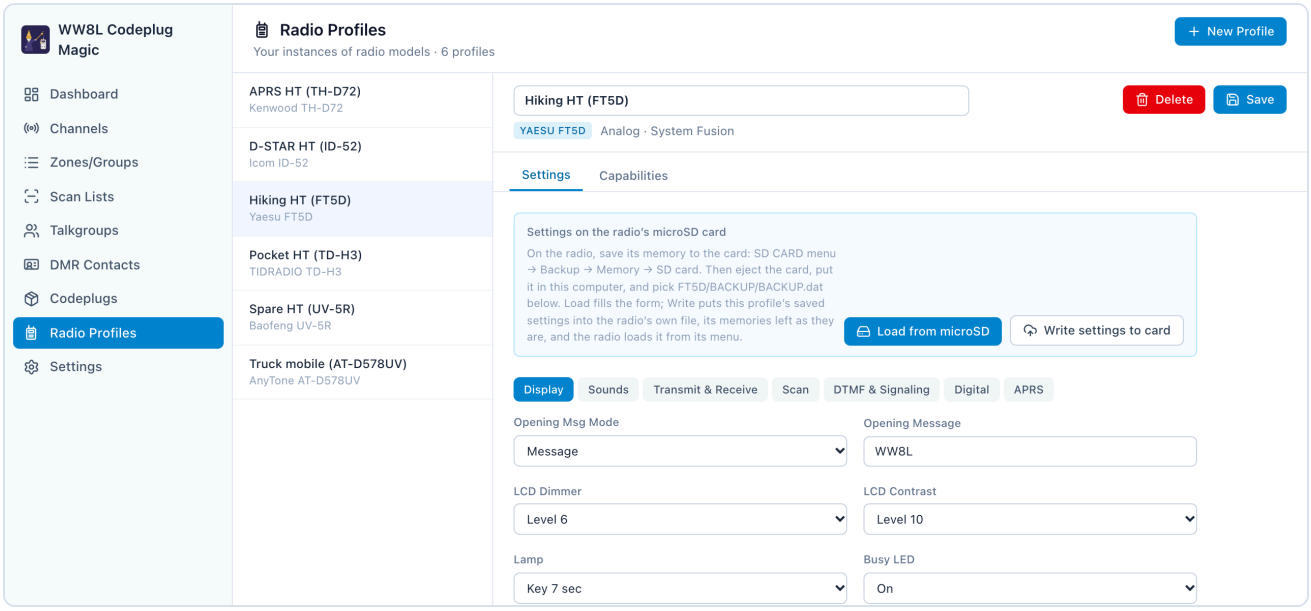
The New Radio Profile dialog. Pick the model; the name is yours.

Fill in the radio's settings

The profile editor's **Settings** tab holds that radio's menu settings, grouped the way the radio groups them. The quickest way to fill it is to read the radio itself:

- **Cable radios:** connect the radio, choose its port, and click **Download from radio**.
- **microSD radios** (FT5D, FTM-300DR, ID-52, TH-D75): save the radio's backup to its card from the radio's own menu. Put the card in the computer and click **Load from microSD**.

Change what you like and click **Save**. The **Capabilities** tab shows what the model supports: memory count, name length, zones or banks, and bands.



A Yaesu FT5D profile. Its settings travel on the radio's own microSD file, so there is a Load from microSD button rather than a cable download.

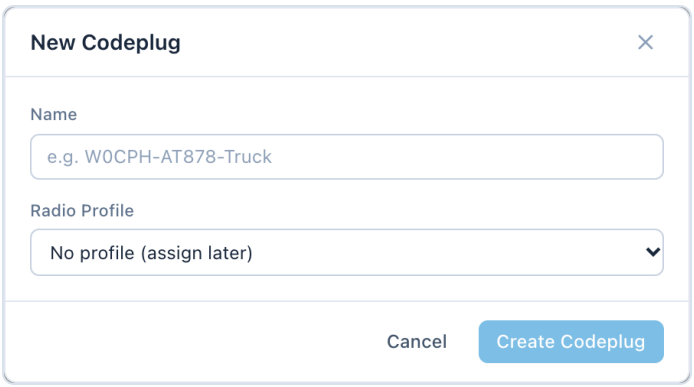
> **A BLANK SETTING MEANS “LEAVE IT ALONE”**
>
> Any setting you leave empty is left as the radio already has it. A brand-new profile with nothing filled in changes nothing on the radio except its channels, so you can't wipe a radio's setup by programming it from a fresh profile.

CHAPTER 6

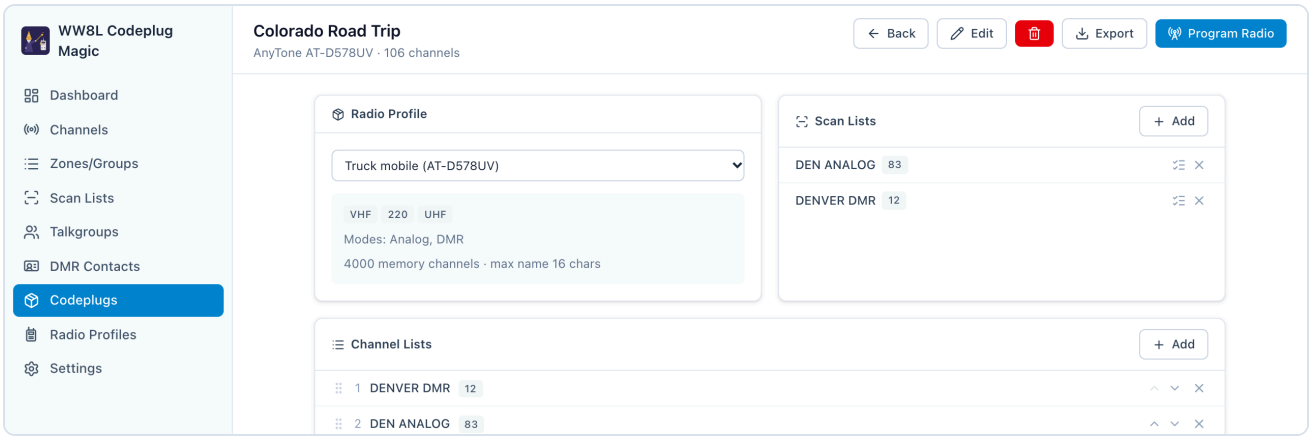
Build a codeplug

A codeplug is a short list of channel lists in the order you want them on the radio. Build it for a purpose (a trip, home, an event), not for a radio.

1. Open **Codeplugs** and click **New Codeplug**.
2. Give it a **Name**. You can pick a **Radio Profile** now or leave it on *No profile (assign later)*. Click **Create Codeplug**.
3. On the codeplug's page, click **Add** on the **Channel Lists** card and tick the lists you want.
4. Drag the lists into order. The first list becomes the radio's first zone or bank, and its channels take the first memories.
5. Optionally, add **Scan Lists**. That card only appears when the selected radio supports scan lists.



The New Codeplug dialog. The profile can wait.



“Colorado Road Trip”: three channel lists (106 channels) and two scan lists, currently pointed at the truck's AT-D578UV.

> **A CHANNEL IN TWO LISTS**
> You can put the same channel in more than one list. On radios where a memory belongs to just one bank or group, it is placed with the *first* list that contains it, so list order also decides that.

CHAPTER 7 · FEATURE

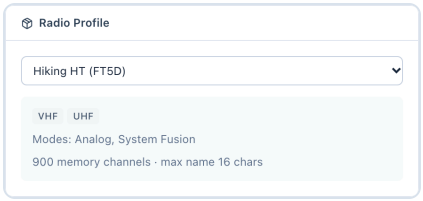
One codeplug, many radios

A codeplug points at one radio profile at a time, and you can change it whenever you like. Build a codeplug once, then program the truck's mobile, your handheld and the spare you lend out from it, switching the profile between each.

1. Open the codeplug. In the **Radio Profile** card, choose the radio you're about to program. It saves immediately and you'll see *Radio profile updated*.
2. Glance at the summary under the dropdown: the radio's bands, its modes, and *memory channels* · *max name* length.
3. Click **Program Radio** (or **Export** for a file). Before anything is written, the dialog shows exactly what this radio will get: how many channels, which are receive-only, which are skipped and why.
4. Program it. Then pick the next radio in the dropdown and repeat.



Truck mobile: VHF, 220 and UHF; analog and DMR; 4000 memories, 16-character names.



Hiking HT: VHF and UHF; analog and System Fusion; 900 memories.



Spare HT: analog only, 128 memories, 7-character names.

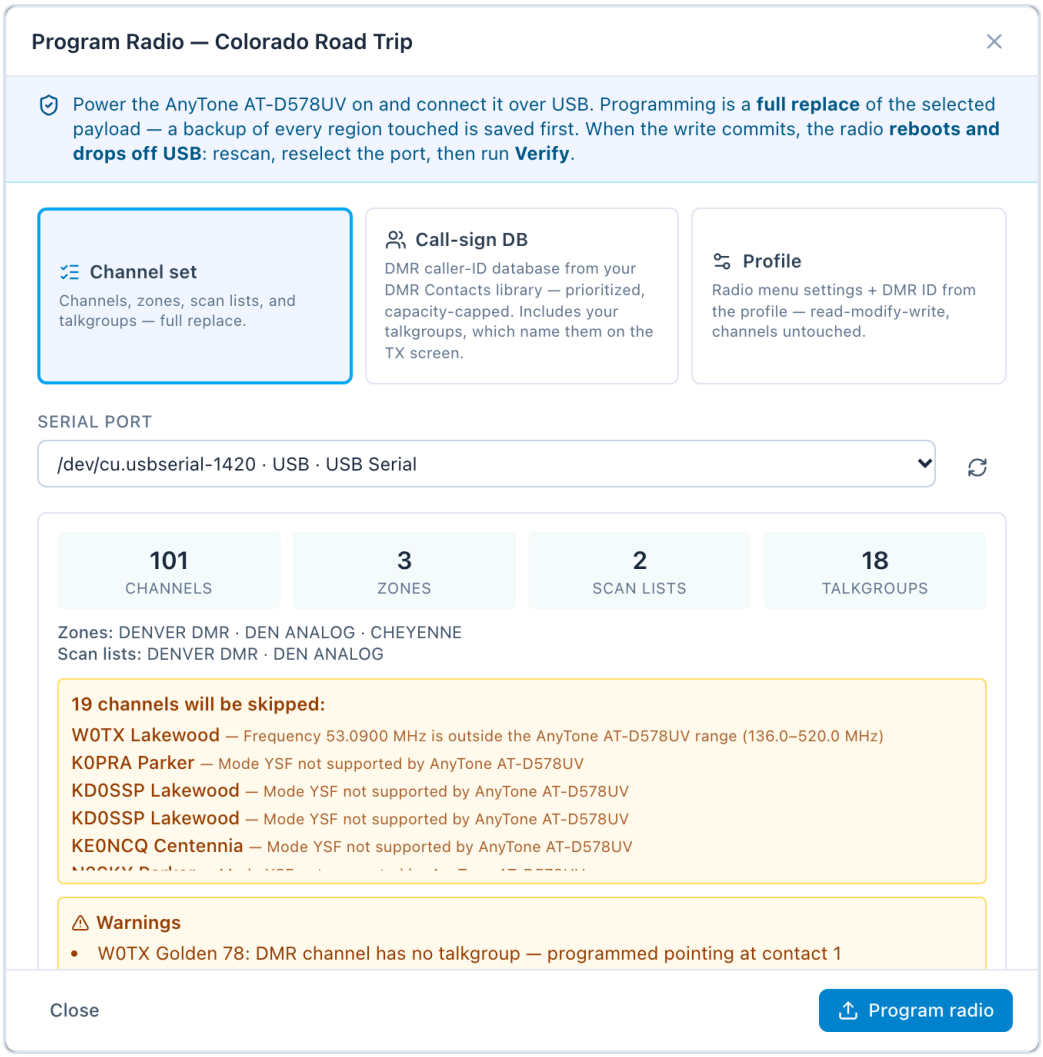
Without a profile, **Export** and **Program Radio** stay disabled. The app needs to know the target radio before it can fit anything to it.

Worked example: one codeplug, three radios

The figures that follow are the real previews for the “Colorado Road Trip” codeplug: 106 channels across three lists, mixing analog FM, DMR, System Fusion, D-STAR and P25 repeaters on 6 m, 2 m, 1.25 m and 70 cm.

1 · Truck mobile: AnyTone AT-D578UV

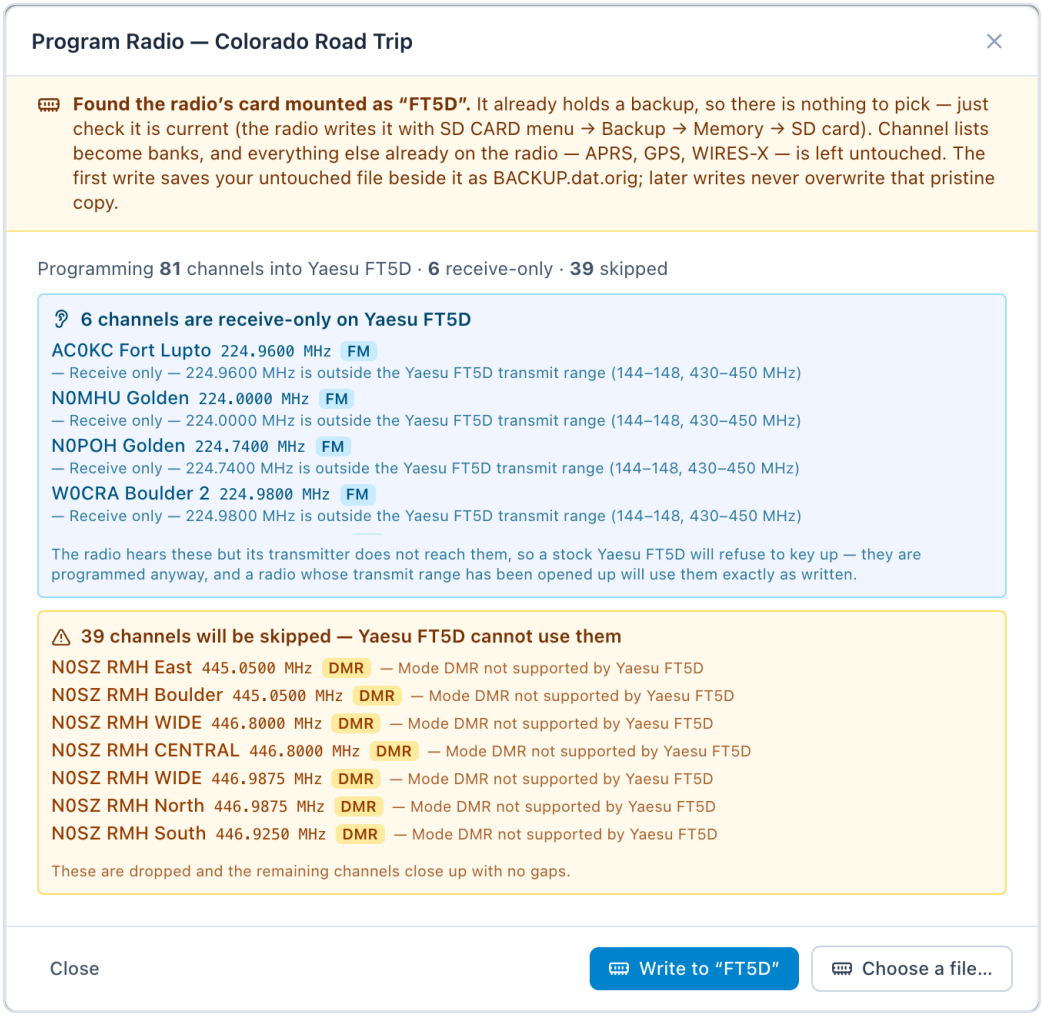
The AnyTone speaks analog and DMR, so DMR repeaters go in. Each one becomes one channel per talkgroup you've assigned to it. The three channel lists become three zones and both scan lists come along. It can't use the System Fusion, D-STAR or P25 repeaters, or the 6 m one below its 136 MHz range, so those 19 are skipped, each with its reason.



AT-D578UV: 101 channels, 3 zones, 2 scan lists, 18 talkgroups; 19 skipped.

2 · Hiking HT: Yaesu FT5D

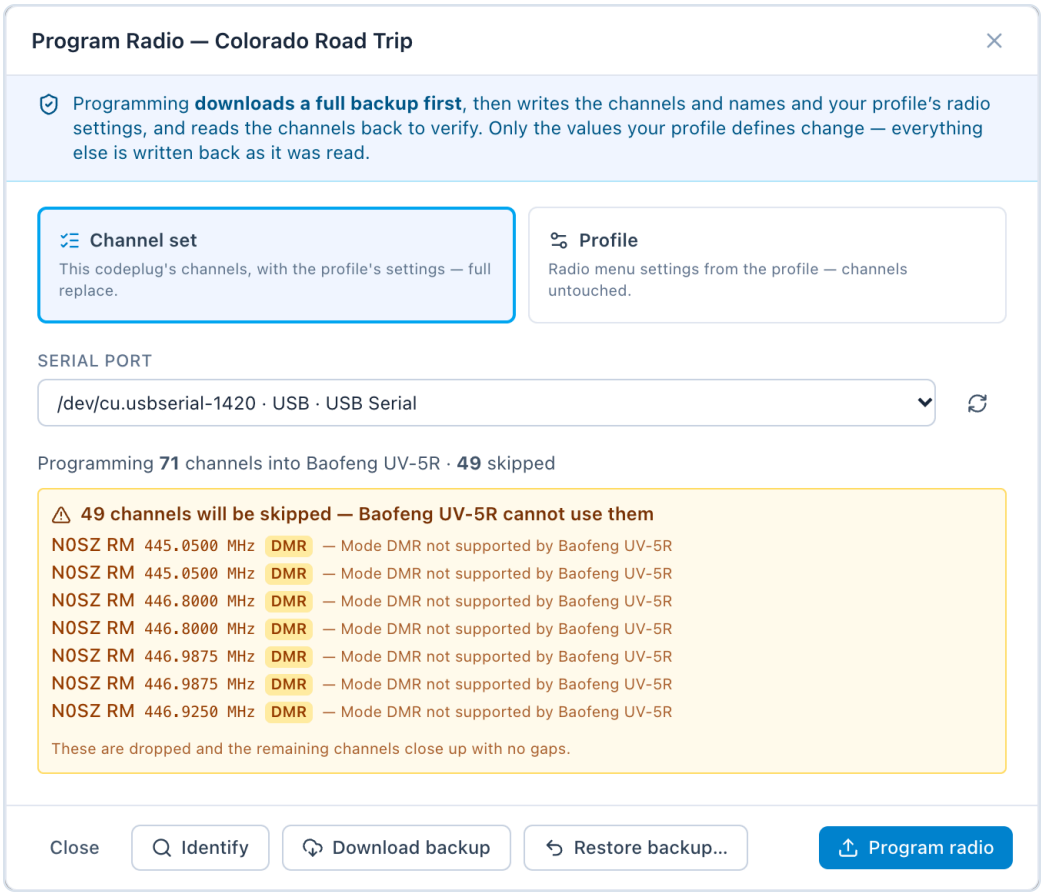
Switch the profile to the FT5D and open Program Radio again. Now System Fusion is in and DMR is out. The FT5D's receiver also covers 6 m and 220 MHz, where its transmitter can't go, so those six channels are programmed **receive-only**: you can listen to them, and the radio won't key up on them. With the card in the computer, the app finds it and offers **Write to “FT5D”**.



FT5D: 81 channels, 6 of them receive-only; 39 skipped. Channel lists become banks.

3 · Spare HT: Baofeng UV-5R

Switch once more. The UV-5R is analog only, so every digital repeater is skipped. Its 7-character display gets the short names: “AC0KC” instead of “AC0KC Fort Lupto”. With no banks, the lists are laid out one after another in a single run of memories.



UV-5R: 71 channels; 49 skipped. Download backup and Restore backup are right there.

The same channels, side by side

CHANNEL	MODE · MHZ	AT-D578UV	FT5D	UV-5R
AC0KC Fort Lupto	FM 224.960	IN AC0KC Fort Lupto	RX ONLY AC0KC Fort Lupto	IN AC0KC
W0TX Lakewood	FM 53.090	~~SKIPPED~~ below 136 MHz	RX ONLY W0TX Lakewood 09	IN W0TX 09
K0PRA Idaho Spri	FM 145.250	IN K0PRA Idaho Sp V	IN K0PRA Idaho Spri	IN K0PRA
N0SZ Boulder 2 talkgroups	DMR 445.050	IN ×2 N0SZ RMH East U N0SZ RMH Boulder	~~SKIPPED~~ no DMR	~~SKIPPED~~ no DMR

Names ending in a letter or number (“V”, “09”) were marked by the app to tell apart two channels that would otherwise have had the same name on that radio. See Chapter 8.

CHAPTER 8

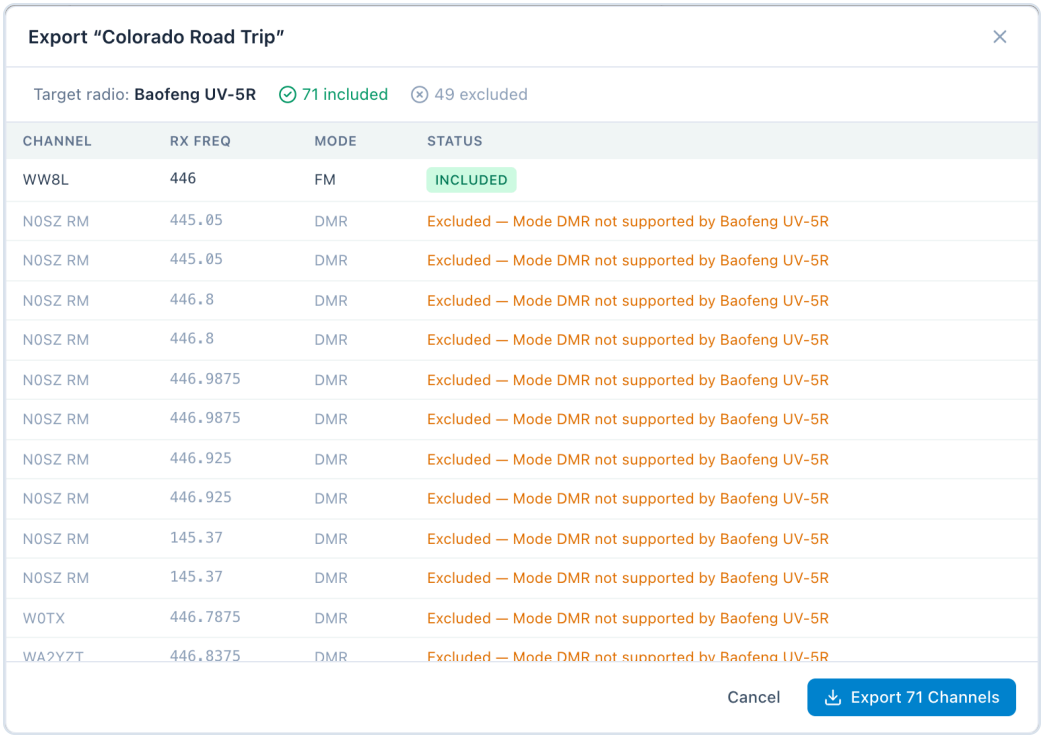
What adapts automatically

Every time you Export or Program Radio, the codeplug is fitted fresh to the profile's radio. Nothing is stored per radio, so switching back and forth never leaves anything stale behind.

SITUATION	WHAT THE APP DOES	WHAT YOU'LL SEE
Mode the radio doesn't have (DMR on an analog radio, Fusion on an AnyTone…)	The channel is skipped.	Listed under *will be skipped*, e.g. “Mode DMR not supported by Yaesu FT5D”.
Radio can hear it but not transmit (GMRS, 220, 6 m, marine, NOAA, air band)	Programmed **receive-only**, not dropped. Where the radio has a per-channel transmit inhibit, it is set.	Listed under *are receive-only* with the radio's transmit range.
Outside the receiver too	Skipped, never written into an empty slot.	“Frequency … is outside the … range”.
Gaps left by skipped channels	The remaining channels close up with no empty memories. On radios with fixed-size zones, they close up within each zone.	“These are dropped and the remaining channels close up with no gaps.”
Channel lists	Become zones (AnyTone, MD-380, BT-9000), banks (FT5D, ID-5100), or memory groups (ID-52, TH-D75). Radios with none get one flat run in list order.	The zone count and zone names in the preview.
DMR repeater with talkgroups	One channel per talkgroup and timeslot, named with the talkgroup. When the long name plus talkgroup won't fit, the short name is used instead.	Repeater “N0SZ Boulder” with talkgroups RMH East and RMH Boulder becomes “N0SZ RMH East” and “N0SZ RMH Boulder”. A DMR channel with no talkgroup is flagged under *Warnings*.
Name length	Uses the long name if it fits; otherwise the short name, then cuts it to the radio's limit.	“AC0KC Fort Lupto” on 16 characters, “AC0KC” on 7.
Two channels, same name	Adds the band letter (U/V…); for two on the same band, the frequency's first two decimals (“09” for 53.09); as a last resort, a counter.	“K0PRA Idaho Sp V”, “W0TX 09”.
Cross-tone channel on a radio without cross-tone	Keeps the transmit tone so you can still open the repeater, and drops the receive side.	The channel is programmed; squelch opens on carrier.
Menu settings	Taken from the profile you selected. Blank settings are left as the radio has them.	On microSD radios the settings ride in the same file as the channels.

Preview before you commit

Export shows the same fit as a table: every channel, its frequency and mode, and whether it's included or excluded and why. Use it to sanity-check a new radio before you plug it in.



The Export dialog for the UV-5R: 71 included, 49 excluded, each with a reason.

CHAPTER 9

How each radio is programmed

The workflow is the same for every radio. Only the last step differs: a cable, the radio's own microSD card, or a file.

RADIO	MODES	CONNECTION	NOTES
Baofeng UV-5R	Analog	USB programming cable	128 memories, flat. Channels and settings.
TIDRADIO TD-H3	Analog	USB programming cable	200 memories. Its Program dialog adds a Radio Options tab.
AnyTone AT-D890UV	DMR + analog	USB	Channels, zones, scan lists, talkgroups, settings, call-sign database.
AnyTone AT-D578UV	DMR + analog	USB	4000 channels in 250 zones; settings incl. APRS; call-sign database.
TYT MD-380	DMR + analog	USB (the radio itself)	Zones of 16, talkgroup contacts, scan lists.
Binteradio BT-9000	Analog	USB programming cable	960 memories in 15 fixed zones. Also sold as Radtel RT-950 Pro.
Kenwood TH-D72	APRS + analog	USB	1000 memories; menu settings.
Kenwood TM-D710	APRS + analog	Serial, operation-panel port	Programmed one memory at a time; settings incl. APRS.
Icom ID-5100	D-STAR + analog	Data cable	1000 memories in named banks A–Z; settings in the same upload.
Yaesu FT5D	Fusion + analog	microSD: `BACKUP.dat`	900 memories in 24 banks; settings.
Yaesu FTM-300DR	Fusion + analog	microSD: `CLNFTM300D.dat`	999 memories, no banks; one M-GRP memory group; settings incl. APRS.
Icom ID-52	D-STAR + analog	microSD: `.icf`	1000 memories in 100 groups; settings.
Kenwood TH-D75	D-STAR + APRS + analog	microSD: `.d75`	1000 memories in 30 groups; settings incl. APRS.
Anything else	—	Export a CHIRP CSV	Program it with CHIRP.

Cable radios

Choose the serial port and click **Program radio**. Every cable write saves a backup of the radio first. Where the radio offers them, **Identify**, **Download backup** and **Restore backup…** sit beside it. AnyTone radios reboot after a write; rescan the port and run **Verify**.

microSD radios

On the radio, save its backup to the card from its SD menu. Put the card in the computer and open Program Radio: the app finds the card, patches the file the radio wrote, and keeps the untouched original beside it. Put the card back and load it from the radio's menu. The dialog shows the exact menu steps for your model.

> **NOTE**
> A microSD file carries the radio's settings as well as its channels, so on these radios the codeplug and the profile travel together in one write.

CHAPTER 10

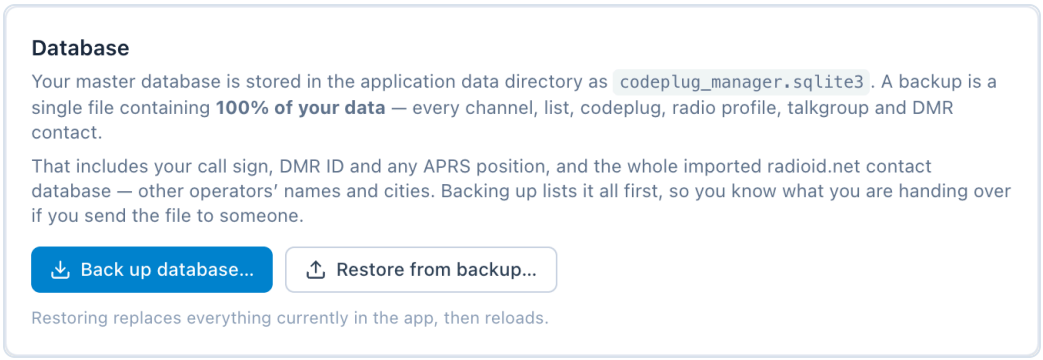
Backing up

Because everything lives in one database, backing up is one file. Radios are backed up too, automatically, every time the app reads or writes them.

Back up the database

Go to **Settings** → **Database** and click **Back up database…**. The backup holds every channel, list, codeplug, radio profile, talkgroup and DMR contact.

Before writing anything, the app lists what the file will contain and flags anything worth thinking about before you share it: your call sign, DMR ID and APRS positions in your radio profiles, and other operators' names from RadioID.net. Then you choose where to save it.



Settings → Database: one file holds all of your data.

Restore from backup… replaces everything currently in the app with the backup. Before it does, it saves your current data to a separate file, so a restore can be undone. This is also how you move to a new computer: back up on the old one, restore on the new one.

Radio backups

Every time the app reads a radio or writes to one, it first saves a copy of that radio's memory. That's what lets you put a radio back exactly as it was. These copies are never deleted for you. Under **Settings** → **Radio backups** you can choose how many to keep per radio and clean up older ones. The copies hold each radio's own data, including your call sign, DMR ID and any stored position, unencrypted, so treat them like the database backup.

CHAPTER 11

Tips

- **Name codeplugs for a purpose, profiles for a radio.** “Colorado Road Trip”, “Home”, “Field Day”; “Truck mobile”, “Hiking HT”. You'll pick codeplug-then-profile without thinking.
- **Read the skipped list before you write.** If a channel you expected is missing, the reason is right there, usually a mode or band the radio doesn't have.
- **Don't trim channels for small radios.** Keep the codeplug complete; the UV-5R simply takes what it can. Trimming by hand means keeping two lists in sync forever.
- **Put the lists you use most first.** On radios with zones or banks they become zone 1 and bank 1, and on flat radios they take the lowest memory numbers.
- **Keep backups.** Cable programming saves one before it writes, and microSD writes keep the radio's original file beside the patched one. On cable radios that offer it, **Download backup** takes a fresh one any time.
- **Lending a radio?** Make a profile for it with the owner's settings, switch the codeplug to it, and program. Your own radios' profiles are untouched.